15.081 Mathematical programming, HW4

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1. **Exercise 5.2:** Sensitivity with respect to changes in basic column A
(Given) LP in standard form, with $A \in \mathbb{R}^{m \times n}$ and m linearly independent rows, optimal basis B, and nondegenerate optimal primal solution, x^*, and dual solution p. Assume first column, A_1 is basic and we change the first entry to $a_{11} \leftarrow a_{11} + \delta$. Lastly define $E \in \mathbb{R}^{m \times m}$ as a matrix of zeros except $e_{11} = 1$.

   (a) **(Show)** for δ small enough, $B + \delta E$ is a basis matrix for the new problem
   $B+\delta E$ is a basis matrix $\iff$ the columns of $B+\delta E$ are linearly independent. **(Contradiction)** Suppose $\exists x \neq 0$ such that

$$\begin{aligned}
(A_1 + \delta e_1)x_1 + \sum_{i \in B(2),\ldots B(m)} A_i x_i &= 0 \\
\delta e_1 x_1 + \sum_{i \in B(1),\ldots B(m)} A_i x_i &= 0 \\
\sum_{i \in B(1),\ldots B(m)} A_i x_i = Bx &= -\delta e_1 x_1 \\
x &= -\delta B^{-1} e_1 x_1
\end{aligned}$$

   If $x_1 = 0 \implies x = 0$ and we have a contradiction. We now consider the case that $x_1 \neq 0$:

$$\begin{aligned}
\tilde{x} &= -\delta B^{-1} e_1 = -\delta [B^{-1}]_1 && (\tilde{x} := x/x_1) \\
\tilde{x}_1 = 1 &= -\delta [B^{-1}]_{11} \\
\delta &= -\frac{1}{[B^{-1}]_{11}}
\end{aligned}$$

   So our assumption that $\exists x \neq 0$ only holds for $\delta = -1/[B^{-1}]_{11}$. Choose any $\delta < -1/[B^{-1}]_{11}$ (or $\delta > -1/[B^{-1}]_{11}$, for that matter) and we do not have the equality $\tilde{x} = -\delta B^{-1} e_1$ and we have a contradiction.

   (b) **(Show)** Under basis $B + \delta E$, the vector $x_B = (I + \delta B^{-1} E)^{-1} B^{-1} b$
   We know

$$\begin{aligned}
(B + \delta E)x_B &= b \\
Bx_B + \delta E x_B &= b \\
x_B + \delta B^{-1} E x_B &= B^{-1} b \\
(I + \delta B^{-1} E)x_B &= B^{-1} b \\
x_B &= (I + \delta B^{-1} E)^{-1} B^{-1} b && (\text{if } (I + \delta B^{-1} E) \text{ invertible})
\end{aligned}$$

   Next we show $(I + \delta B^{-1} E)$ is invertible. Note by rearranging that $(B + \delta E) = B(I + \delta B^{-1} E)$. Since B invertible and $(B + \delta E)$ invertible (from part (a)), we must also have that $(I + \delta B^{-1} E)$ invertible.

(c) **(Show)** for δ small enough, $B + \delta E$ is an optimal basis matrix for the new problem
$B+\delta E$ is an optimal basis matrix $\implies x_B \geq 0$ (perturbed problem feasibility) and $\tilde{\bar{c}} \geq 0$ (perturbed problem optimality)

i. $(x_B \geq 0)$
B optimal basis $\implies x_B^* = B^{-1}b \geq 0$. And from part (b) we've shown that $(B + \delta E)^{-1}b = (I + \delta B^{-1}E)^{-1}B^{-1}b$. Observe the form of $(I + \delta B^{-1}E)$ can be seen as a rank-1 update to an invertible matrix

$$(I + \delta B^{-1}E) = (I + \delta B_1^{-1}e_1^\top)$$

Using the Sherman-Morrison formula for matrix inverses

$$\begin{aligned}(I + \delta B_1^{-1}e_1^\top)^{-1} &= I^{-1} - \frac{1}{1 + e_1^\top I^{-1}\delta B_1^{-1}} I^{-1}(\delta B_1^{-1}e_1^\top)I^{-1} \\ &= I - \frac{1}{1 + \delta e_1^\top B_1^{-1}}(\delta B_1^{-1}e_1^\top) \\ &= I - \frac{\delta}{1 + \delta B_{11}^{-1}} B^{-1}E\end{aligned}$$

And

$$\begin{aligned}x_B(\delta) &= (I - \frac{\delta}{1 + \delta B_{11}^{-1}} B^{-1}E)B^{-1}b \\ &= B^{-1}b - \frac{\delta}{1 + \delta B_{11}^{-1}} B^{-1}EB^{-1}b \\ &= B^{-1}b - \gamma(\delta)B^{-1}EB^{-1}b && (\text{for } \gamma(\delta) = \tfrac{\delta}{1+\delta B_{11}^{-1}}, \delta \neq -\tfrac{1}{B_{11}^{-1}}) \\ &= B^{-1}b - \gamma(\delta)v && (\text{for } v = f(B^{-1}, b) = B^{-1}EB^{-1}b)\end{aligned}$$

Thus

$$x_B(\delta) \;=\; x_B^* - \gamma(\delta)\, v, \quad x_B^* := B^{-1}b \succeq 0, \quad \gamma(\delta) := \frac{\delta}{1 + \delta B_{11}^{-1}}, \quad v := B^{-1}EB^{-1}b.$$

For each coordinate i,

$$[x_B(\delta)]_i = [x_B^*]_i - \gamma(\delta)\, v_i.$$

- If $v_i > 0$, this is nonnegative provided $\gamma(\delta) \leq [x_B^*]_i / v_i$.
- If $v_i < 0$, it is nonnegative provided $\gamma(\delta) \geq [x_B^*]_i / v_i$.
- If $v_i = 0$, there is no restriction.

Thus we may choose

$$\gamma(\delta) = \frac{\delta}{1 + \delta B_{11}^{-1}}, \quad \delta \neq -\frac{1}{B_{11}^{-1}}, \quad \text{and} \quad |\gamma(\delta)| \leq \min_{i: v_i \neq 0} \frac{[B^{-1}b]_i}{|v_i|},$$

Note $\exists \gamma(\delta)$ since $\delta \to 0 \implies \gamma(\delta) \to 0$ so we can choose sufficiently small $\gamma(\delta)$ for all requirements to hold.

ii. $(\tilde{\bar{c}} \geq 0)$
Under the original problem, x_B^* optimal and $\bar{c} = c - c_B^\top B^{-1}A \geq 0$. Now under the perturbed problem we have

$$\begin{aligned}\tilde{\bar{c}} = [\tilde{\bar{c}}_B, \tilde{\bar{c}}_N] &= \left[c_B - c_B^\top (B + \delta E)^{-1}(B + \delta E), c_N - c_B^\top (B + \delta E)^{-1}N\right] \\ [\tilde{\bar{c}}_B, \tilde{\bar{c}}_N] &= \left[c_B - c_B I, c_N - c_B^\top (B + \delta E)^{-1}N\right]\end{aligned}$$

Since $\bar{c}_B = 0$ in the new perturbed problem, suffice to confirm $\tilde{\bar{c}}_N \geq 0$:

$$\begin{aligned}
\tilde{\bar{c}}_N &= c_N - c_B^\top (B+\delta E)^{-1} N \\
&= c_N - c_B^\top (B^{-1} - \gamma(\delta) B^{-1} E B^{-1}) N && \text{(for } \gamma(\delta) = \tfrac{\delta}{1+\delta B_{11}^{-1}}, \delta \neq -\tfrac{1}{B_{11}^{-1}}\text{)} \\
&= \left(c_N - c_B^\top B^{-1} N\right) - \gamma(\delta) c_B^\top B^{-1} E B^{-1} N \\
&= \bar{c}_N - \gamma(\delta) u && \text{(for } \bar{c}_N = c_N - c_B^\top B^{-1} N, u = c_B^\top B^{-1} E B^{-1} N\text{)}
\end{aligned}$$

For each coordinate i,

$$\tilde{\bar{c}}_{N(i)} = \bar{c}_{N(i)} - \gamma(\delta) u_i$$

- If $u_i > 0$, this is nonnegative provided $\gamma(\delta) \leq \bar{c}_{N(i)}/u_i$.
- If $u_i < 0$, it is nonnegative provided $\gamma(\delta) \geq \bar{c}_{N(i)}/u_i$.
- If $u_i = 0$, there is no restriction.

Thus we may choose

$$\gamma(\delta) = \frac{\delta}{1+\delta B_{11}^{-1}}, \quad \delta \neq -\frac{1}{B_{11}^{-1}}, \quad \text{and} \quad |\gamma(\delta)| \leq \min_{i:u_i \neq 0} \frac{|\bar{c}_{N(i)}|}{|u_i|}$$

Because $\gamma(\delta) \to 0$ as $\delta \to 0$, $\exists \delta_0$ such that $\forall |\delta| < \delta_0$, both conditions hold and $\delta \neq -1/B_{11}^{-1}$

(d) **(Given)** $(I + \delta B^{-1} E)^{-1} \approx I - \delta B^{-1} E$, where $\approx$ indicates that second-order terms are ignored.
(Show) $c_B^\top x_B \approx c^\top x^* - \delta p_1 x_1^*$, where x_B is defined in part (b).

$$\begin{aligned}
x_B &= (I - \delta B^{-1} E)^{-1} B^{-1} b \\
x_B &\approx (I - \delta B^{-1} E) B^{-1} b = B^{-1} b - \delta B^{-1} E B^{-1} b \\
c_B^\top x_B &\approx c_B^\top B^{-1} b - \delta c_B^\top B^{-1} E B^{-1} b \\
c_B^\top x_B &\approx c_B^\top x_B^* - \delta c_B^\top B^{-1} E x_B^* && \text{(since } B^{-1} b = x_B^*\text{)} \\
c_B^\top x_B &\approx c^\top x^* - \delta p^\top E x_B^* && \text{(since } c_B^\top x_B^* = c^\top x^*, p^\top = c_B^\top B^{-1}\text{)} \\
c_B^\top x_B &\approx c^\top x^* - \delta p_1 e_1^\top x_B^* && \text{(since } p^T E = p_1 e_1^\top\text{)} \\
c_B^\top x_B &\approx c^\top x^* - \delta p_1 x_1^* && \text{(since } e_1^\top x_B^* = x_{B(1)}^* = x_1^* \text{ for } A_1 \text{ basic)}
\end{aligned}$$

2. **Exercise 5.3:** Sensitivity with respect to changes in basic column A
(Given) LP in standard form, with $A \in \mathbb{R}^{m \times n}$ and m linearly independent rows. Assume $A_1, \ldots, A_m$ form optimal basis, B, and suppose we change $A_1 \leftarrow A_1 + \delta A_0$. Define $B(\delta) = [A_1 + \delta A_0, A_2, \ldots, A_m]$, and $[\delta_1, \delta_2]$ to be the closed interval range of values of δ for which $\det B(\delta) \neq 0$.
(Show) The subset of $[\delta_1, \delta_2]$ for which $B(\delta)$ is an optimal basis is a closed interval.

We begin by characterizing when $B(\delta)$ is an optimal basis and then analyzing the resulting set of feasible δ.

$$B(\delta) \text{ an optimal basis } \implies \begin{cases} x_B(\delta) \geq 0 & x_B(\delta) = (B(\delta))^{-1} b \geq 0 \\ \bar{c}(\delta) \geq 0 & \bar{c}(\delta) = c - c_{B(\delta)}^\top (B(\delta))^{-1} A(\delta) \end{cases}$$

Inverting $B(\delta)$ using Sherman-Morrison formula. Note $B(\delta) = B + \delta A_0 e_1^\top$. Making the same observation as in Exercise 5.3.b,

$$B(\delta) = B + \delta A_0 e_1^\top = (I + \delta A_0 e_1^\top B^{-1}) B \qquad B(\delta)^{-1} = (B + \delta A_0 e_1^\top)^{-1} = B^{-1} (I + \delta A_0 e_1^\top B^{-1})^{-1}$$

When $(I + \delta A_0 e_1^\top B^{-1})$ is invertible. As a given, we know $B(\delta)$ invertible for $\delta \in [\delta_1, \delta_2]$, and therefore this term is invertible within the same δ-range.

Taking the same steps as in Exercise 5.3.c, we decompose the inverse of $(I + \delta A_0 e_1^\top B^{-1})$ using the Sherman-Morrison formula for matrix inversions. First note the rank-1 update form of the matrix

$$(I + \delta A_0 e_1^\top B^{-1}) \iff (I + \delta A_0 b_1^{-1\top}) \qquad \text{(where } b_1^{-1\top} \text{ is row 1 of } B^{-1})$$

Then take the inversion decomposition

$$\begin{aligned}(I + \delta A_0 b_1^{-1\top})^{-1} &= I^{-1} - \frac{\delta}{1 + \delta b_1^{-1\top} I^{-1} A_0} I^{-1} A_0 b_1^{-1\top} I^{-1} \\ &= I - \frac{\delta}{1 + \delta b_1^{-1\top} A_0} A_0 b_1^{-1\top} \qquad \text{(for } 1 + \delta b_1^{-1\top} A_0 \neq 0)\end{aligned}$$

Feasible δ for Sherman-Morrison formula We see that $(1 + \delta b_1^{-1\top} A_0) \neq 0$ for $\delta \in [\delta_1, \delta_2]$ since

$$\begin{aligned}\det(B(\delta)) &= \det((I + \delta A_0 b_1^{-1\top})B) = \det(I + \delta A_0 b_1^{-1\top}) \det(B) \\ &= \det(B)(1 + \delta b_1^{-1\top} A_0) \qquad \text{(since } \det(I + uv^\top) = 1 + v^\top u) \\ &\neq 0, \text{ for } \delta \in [\delta_1, \delta_2] \text{ from given}\end{aligned}$$

With this setup, we are prepared to define the set of δ for which $x_B(\delta) \geq 0$ (perturbed problem feasibility) and $\bar{c}_B(\delta) \geq 0$ (perturbed problem optimality).

Feasible δ for $x_B(\delta) \geq 0$.

$$\begin{aligned}x_B(\delta) &= B^{-1}(I + \delta A_0 e_1^\top B^{-1})^{-1} b \\ &= B^{-1}\left(I - \frac{\delta}{1 + \delta b_1^{-1\top} A_0} A_0 b_1^{-1\top}\right) b = B^{-1} b - \frac{\delta}{1 + \delta b_1^{-1\top} A_0} B^{-1} A_0 b_1^{-1\top} b\end{aligned}$$

For feasibility of the perturbed problem, we need to choose $\delta \in [\delta_1, \delta_2]$ such that

$$\begin{aligned}B^{-1} b &\succeq \frac{\delta}{1 + \delta b_1^{-1\top} A_0} B^{-1} A_0 b_1^{-1\top} b \\ B^{-1} b &\succeq \gamma(\delta) p \qquad \text{(for scalar } \gamma(\delta) = \delta/(1 + \delta b_1^{-1\top} A_0) \text{ and vector } p = B^{-1} A_0 b_1^{-1\top} b) \\ [B^{-1} b]_i &\geq \gamma(\delta) p_i \ \forall i \in [m] \\ \gamma(\delta) &\in \left[\max_{\{i : p_i < 0\}} [B^{-1} b]_i / p_i, \min_{\{i : p_i > 0\}} [B^{-1} b]_i / p_i\right] := [\underline{\gamma}_p, \overline{\gamma}_p] \\ &\qquad \text{(unbounded on one side if } \{i : p_i \overset{<}{>} 0\} = \emptyset)\end{aligned}$$

Note $\gamma(\delta) \in [\underline{\gamma}_p, \overline{\gamma}_p]$ is a continuous function of the form $\delta/(1 + \alpha\delta)$. This function increases monotonically over any interval that does not include $\delta : 1 + \alpha\delta = 0$. Since we are given that $(1 + \delta b_1^{-1\top} A_0) \neq 0$ over $[\delta_1, \delta_2]$, we have that

$$\gamma(\delta) \in [\underline{\gamma}_p, \overline{\gamma}_p] \Longrightarrow \delta \in [\delta_{1_{\gamma p}}, \delta_{2_{\gamma p}}]$$

In words, the set of feasible δ for $x_B(\delta) \geq 0$ is an interval, which may be open on either side if $\{i : p_i \overset{<}{>} 0\} = \emptyset$.

Feasible δ for $\bar{c}_N(\delta) \geq 0$.

$$\bar{c}(\delta) = [\bar{c_B}(\delta), \bar{c_N}(\delta)] = \left[c_B - c_B^\top (B(\delta))^{-1} B(\delta), c_N - c_B^\top (B(\delta))^{-1} N\right]$$
$$[\bar{c_B}(\delta), \bar{c_N}(\delta)] = \left[c_B - c_B I, c_N - c_B^\top (B(\delta))^{-1} N\right]$$

Since $\bar{c}_B(\delta) = 0$ in the new perturbed problem, suffice to confirm $\bar{c_N}(\delta) \geq 0$:

$$\begin{aligned}
\bar{c_N}(\delta) &= c_N - c_B^\top (B + \delta A_0 e_1^\top)^{-1} N = c_N - c_B^\top B^{-1} (I + \delta A_0 e_1^\top B^{-1})^{-1} N \\
&= c_N - c_B^\top B^{-1} \left(I - \frac{\delta}{1 + \delta b_1^{-1\top} A_0} A_0 b_1^{-1\top} \right) N \\
&= c_N - c_B^\top B^{-1} N - \frac{\delta}{1 + \delta b_1^{-1\top} A_0} c_B^\top B^{-1} A_0 b_1^{-1\top} N
\end{aligned}$$

Note $c_N - c_B^\top B^{-1} = \bar{c}_N$ from the original problem for notation simplicity. For optimality of the perturbed problem, we need to choose $\delta \in [\delta_1, \delta_2]$ such that

$$\begin{aligned}
\bar{c}_N &\succeq \frac{\delta}{1 + \delta b_1^{-1\top} A_0} c_B^\top B^{-1} A_0 b_1^{-1\top} N \\
\bar{c}_N &\succeq \gamma(\delta) q \qquad \text{(for scalar } \gamma(\delta) \text{ defined above and vector } q = c_B^\top B^{-1} A_0 b_1^{-1\top} N) \\
[\bar{c}_N]_i &\geq \gamma(\delta) q_i \ \forall i \in [m] \\
\gamma(\delta) &\in \left[\max_{\{i: q_i < 0\}} [\bar{c}_N]_i / q_i, \min_{\{i: q_i > 0\}} [\bar{c}_N]_i / q_i \right] := [\underline{\gamma}_q, \overline{\gamma}_q] \qquad \text{(unbounded on one side if } \{i : q_i \overset{<}{>} 0\} = \emptyset)
\end{aligned}$$

Using the same logic as above about monotonically increasing $\gamma(\delta)$ over $[\delta_1, \delta_2]$, we have that

$$\gamma(\delta) \in [\underline{\gamma}_q, \overline{\gamma}_q] \implies \delta \in [\delta_{1_{\gamma q}}, \delta_{2_{\gamma q}}]$$

Feasible δ is a closed interval From the requirements presented above, feasible δ must be the intersection of closed interval, $[\delta_1, \delta_2]$, with interval $[\delta_{1_{\gamma p}}, \delta_{2_{\gamma p}}]$ to guarantee feasibility of $x_B(\delta)$ and interval $[\delta_{1_{\gamma q}}, \delta_{2_{\gamma q}}]$ to guarantee optimality of $\bar{c}(\delta)$.

$$\delta \in [\delta_1, \delta_2] \cap [\delta_{1_{\gamma p}}, \delta_{2_{\gamma p}}] \cap [\delta_{1_{\gamma q}}, \delta_{2_{\gamma q}}]$$

The intersection of a closed interval with two other intervals (open or closed) is a closed interval, hence we have proved the result.

3. **Exercise 5.14**
(Given) Given LP

$$\begin{aligned}
\min_x \quad & (c + \theta d)^\top x \\
\text{s.t.} \quad & Ax = b + \theta f \\
& x \geq 0
\end{aligned}$$

Where $A \in \mathbb{R}^{m \times n}$ with m linearly independent rows. Assume the problem is feasible and optimal cost $f(\theta)$ finite for $\theta \in [\theta_1, \theta_2]$.

(a) **(Given)** A basis B is optimal for $\theta = -10$ and $\theta = 10$. **(Prove)** B is optimal for $\theta = 5$.

i. ($\theta = 5$ feasible) The provided θ values for optimal basis B tell us

$$x_{B_{10}} = B^{-1}(b + 10f) \succeq 0 \qquad\qquad x_{B_{-10}} = B^{-1}(b - 10f) \succeq 0$$

Observe that

$$x_{B_5} = B^{-1}(b + 5f) = \frac{3}{4}\underbrace{B^{-1}(b + 10f)}_{\succeq 0} + \frac{1}{4}\underbrace{B^{-1}(b - 10f)}_{\succeq 0} \succeq 0$$

So basis B feasible for $\theta = 5$

ii. ($\theta = 5$ optimal) The provided θ values for optimal basis B tell us

$$\bar{c}_{10} = (c + 10d) - c_B^\top B^{-1}A \succeq 0 \qquad\qquad \bar{c}_{-10} = (c - 10d) - c_B^\top B^{-1}A \succeq 0$$

Observe that

$$\bar{c}_5 = (c + 5d) - c_B^\top B^{-1}A = \frac{3}{4}\underbrace{(c + 10d) - c_B^\top B^{-1}A}_{\succeq 0} + \frac{1}{4}\underbrace{(c - 10d) - c_B^\top B^{-1}A}_{\succeq 0} \succeq 0$$

So basis B optimal for $\theta = 5$

(b) **(Show)** $f(\theta)$ is piecewise quadratic function of θ and give an upper bound on the number of pieces

We are given that this LP is feasible within a range of θ. Feasibility tells us that within this range, $\exists B(\theta)$ such that

$$x^*_{B(\theta)} = B(\theta)^{-1}(b + \theta f)$$

The optimal cost can be written as

$$\begin{aligned} f(\theta) = (c + \theta d)^\top x^* &= (c + \theta d)^\top_{B(\theta)} x^*_{B(\theta)} \\ &= c^\top_{B(\theta)} x^*_{B(\theta)} + \theta d^\top_{B(\theta)} x^*_{B(\theta)} \\ &= c^\top_{B(\theta)} B(\theta)^{-1}(b + \theta f) + \theta d^\top_{B(\theta)} B(\theta)^{-1}(b + \theta f) \\ &= \theta^2 d^\top_{B(\theta)} B(\theta)^{-1} f \;+\; \theta\left(c^\top_{B(\theta)} B(\theta)^{-1} f + d^\top_{B(\theta)} B(\theta)^{-1} b\right) \;+\; c^\top_{B(\theta)} B(\theta)^{-1} b \end{aligned}$$

Which, for every fixed optimal basis, $B(\theta)$, is a quadratic function of θ since it can be written in the form

$$f(\theta) = \theta^2 v_{B(\theta)} + \theta u_{B(\theta)} + t_{B(\theta)}$$

Here we notice that each optimal basis may define a different quadratic function in θ and for $A \in \mathbb{R}^{m \times n}$ with m linearly independent constraints, we have $\binom{n}{m}$ possible bases corresponding to distinct quadratic equations.

(c) **(Given)** $b = 0, c = 0$ and a certain basis is optimal for $\theta = 1$. **(Answer)** For what other nonnegative values of θ is the same basis optimal?

Rewriting the LP with new givens, we have

$$\begin{aligned} \min_x \quad & \theta d^\top x \\ \text{s.t.} \quad & Ax = \theta f \\ & x \geq 0 \end{aligned}$$

Let B be the optimal basis for $\theta = 1$. We know $B^{-1}f \succeq 0$ and $d - d_B^T B^{-1}A \succeq 0$. To determine the range of values of theta where B remains optimal, consider

$$\theta B^{-1} f \succeq 0 \qquad\qquad \theta d - \theta d_B^T B^{-1} A \succeq 0$$

Feasibility Clearly all $\theta \geq 0$ are feasible since this multiplies a nonnegative number ($\theta \geq 0$) by a nonnegative vector ($B^{-1}f \succeq 0$).

Optimality B remains optimal so long as $\theta d - \theta d_B^\top B^{-1}A \succeq 0$. And we observe

$$\theta d - \theta d_B^\top B^{-1}A = \theta \underbrace{(d - d_B^\top B^{-1}A)}_{\succeq 0} \succeq 0 \qquad \text{(for } \theta \geq 0\text{)}$$

Together, we've shown all $\theta \geq 0$ preserve feasibility and optimality of optimal basis B.

(d) Is $f(\theta)$ convex, concave, or neither?
In part (b) we showed that $f(\theta)$ has the form $f(\theta) = \theta^2 v_{B(\theta)} + \theta u_{B(\theta)} + t_{B(\theta)}$ for varying $(v_{B(\theta)}, u_{B(\theta)}, t_{B(\theta)})$ depending on optimal basis $B(\theta)$. $f(\theta)$ is convex (concave) if $v_{B(\theta)} \geq 0$ ($v_{B(\theta)} \leq 0$) for every optimal basis, $B(\theta)$, and it is neither if the piecewise quadratic equations are not all consistently ≥ 0 or ≤ 0
Now considering $v_{B(\theta)} = d_{B(\theta)}^\top B(\theta)^{-1} f$, there is nothing in the problem data, (d, B, f) requiring $v_{B(\theta)}$ to be either strictly positive or strictly negative, so without more restrictions on $f(\theta)$ it is neither guaranteed to be convex nor concave, so it is most likely neither.

4. **Exercise 6.3 (Given)** The LP in the text book

(a) **(Given)** Two feasible solutions are provided below. Construct a restricted master problem (RMP) in which x is constrained to be a convex combination of x^1 and x^2. **(Also want)** The optimal primal solution and the corresponding simplex multipliers (dual variables).

$$x^1 = (x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}) = (20, 0, 0, 0, 10, 10)$$
$$x^2 = (x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}) = (0, 10, 10, 20, 0, 0)$$

The cost and linking vectors are

$$c = [0, 1, 0, 0, 1, 1]^\top, \qquad d = [1, 0, 0, 0, 0, 1].$$

The restricted master problem uses the coefficients of each column x^i:

$$\begin{aligned} \max_{\lambda} \quad & \lambda_1 c^\top x^1 + \lambda_2 c^\top x^2 \\ \text{s.t.} \quad & \lambda_1 d^\top x^1 + \lambda_2 d^\top x^2 \leq 15 && \text{(linking constraint)} \\ & \lambda_1 + \lambda_2 = 1, \quad \lambda_i \geq 0 && \text{(convexity constraint.)} \end{aligned}$$

Plugging in numbers:

$$\begin{aligned} \max_{\lambda} \quad & 20\lambda_1 + 10\lambda_2 \\ \text{s.t.} \quad & 30\lambda_1 + 0\lambda_2 \leq 15 \\ & \lambda_1 + \lambda_2 = 1, \quad \lambda_i \geq 0. \end{aligned}$$

By inspection, the resource constraint is binding and both columns are active at optimality:

$$\lambda_1 = \lambda_2 = 0.5, \qquad z^* = 20(0.5) + 10(0.5) = 15.$$

Finding the simplex multipliers (dual variables). Let r be the dual on the convexity constraint (free), and let q be the dual variable on the linking constraint ($q \geq 0$). The Lagrangian of is

$$L(\lambda, q, r) = 20\lambda_1 + 10\lambda_2 + r(1 - \lambda_1 - \lambda_2) + q(15 - 30\lambda_1), \quad q \geq 0.$$

Maximizing L with respect to $\lambda \geq 0$ gives

$$g(q, r) = r + 15q + \max_{\lambda \geq 0} \{(20 - r - 30q)\lambda_1 + (10 - r)\lambda_2\}.$$

For the maximization to be finite, both coefficients of λ_1 and λ_2 must be nonpositive, so the dual feasible region is

$$\mathcal{P} = \{(q, r) \mid 20 - r - 30q \leq 0,\ 10 - r \leq 0,\ q \geq 0\}.$$

And the dual problem is

$$\begin{aligned} \min_{q,r} \quad & r + 15q \\ \text{s.t.} \quad & r + 30q \geq 20, \\ & r \geq 10, \\ & q \geq 0 \end{aligned}$$

By inspection, the dual is minimized at $r = 10,\ q = \frac{1}{3}$, the simplex multipliers.

(b) **(Want)** Using the simplex multipliers $(q, r) = (\frac{1}{3}, 10)$ from part (a), formulate the subproblem and solve by inspection. The subproblem is defined as

$$\begin{aligned} \max_{x} \quad & (c - qd)^\top x && (q \text{ is the dual on the linking constraint}) \\ \text{s.t.} \quad & Ax = b && (\text{the uncoupled constraints defining } P) \\ & x \geq 0 \end{aligned}$$

Substituting numbers:

$$\begin{aligned} \max_{x} \quad & [-\tfrac{1}{3}, 1, 0, 0, 1, \tfrac{2}{3}]\, x \\ \text{s.t.} \quad & x_{11} + x_{12} + x_{13} = 20 \\ & x_{21} + x_{22} + x_{23} = 20 \\ & x_{11} + x_{21} = 20 \\ & x_{12} + x_{22} = 10 \\ & x_{13} + x_{23} = 10 \\ & x \geq 0 \end{aligned}$$

Eliminating the dependent variables x_{21}, x_{22}, x_{23}:

$$x_{21} = 20 - x_{11}, \qquad x_{22} = 10 - x_{12}, \qquad x_{23} = 10 - x_{13}$$

Substitute and simplify:

$$\begin{aligned} \max_{x_{11}, x_{13}} \quad & -\tfrac{1}{3}x_{11} - \tfrac{2}{3}x_{13} + \tfrac{50}{3} \\ \text{s.t.} \quad & x_{11} + x_{12} + x_{13} = 20, \\ & 0 \leq x_{11} \leq 20,\ 0 \leq x_{12} \leq 10,\ 0 \leq x_{13} \leq 10 \end{aligned}$$

To maximize the objective we push $x_{11} + x_{13}$ to its smallest feasible value (10), and assign that total to the variable with the smaller penalty (x_{11}).

Therefore:

$$\begin{aligned} x_{11} = 10, \quad x_{13} = 0, \quad x_{12} = 10 \\ x_{21} = 10, \quad x_{22} = 0, \quad x_{23} = 10 \qquad \text{(backsubstitute)} \\ x^3 = [10, 10, 0, 10, 0, 10] \end{aligned}$$

The corresponding objective value is

$$\begin{aligned} (c - qd)^\top x^3 &= [-\tfrac{1}{3}, 1, 0, 0, 1, \tfrac{2}{3}]^\top [10, 10, 0, 10, 0, 10] \\ &= -\tfrac{10}{3} + 10 + \tfrac{20}{3} = \tfrac{40}{3} \\ \tfrac{40}{3} > r = 10 \Longrightarrow (c - qd)^\top x^3 - r > 0 \end{aligned}$$

which means the subproblem produced an improving column (positive reduced cost in a maximization context), and x^3 should be added to the restricted master problem.

(c) **(Want)** What is the reduced cost variable, λ_i, associated with the optimal extreme point x^i obtained from the subproblem solved in part (b)?

After adding x^3 as a new column, the restricted master problem becomes:

$$\begin{aligned} \max_{\lambda} \quad & 20\lambda_1 + 10\lambda_2 + 20\lambda_3 \\ \text{s.t.} \quad & 30\lambda_1 + 0\lambda_2 + 20\lambda_3 \leq 15 && \text{(linking constraint)} \\ & \lambda_1 + \lambda_2 + \lambda_3 = 1, \quad \lambda_i \geq 0. && \text{(convexity constraint)} \end{aligned}$$

By visual inspection of the convex combinations, this problem attains the same optimal solution as in part (a): $\lambda_1 = \lambda_2 = 0.5, \quad \lambda_3 = 0$ and the objective remains $z^* = 15$

(d) **(Want)** An upper bound on the optimal cost

Consider the Dantzig-Wolfe formulation of the master problem and the restricted master problem

$$\begin{aligned} P^{(MP)}: \max_{\lambda} \quad & \sum_{i \in \mathcal{I}} \lambda_i c^\top x^i && \Longleftrightarrow && \max_{\lambda} \quad c^\top X \lambda \\ \text{s.t.} \quad & \sum_{i \in \mathcal{I}} \lambda_i d^\top x^i = b && && \text{s.t.} \quad d^\top X \lambda = b \\ & \sum_{i \in \mathcal{I}} \lambda_i = 1, \ \lambda_i \geq 0 && && \lambda^\top \mathbf{1} = 1, \ \lambda \geq 0 \end{aligned}$$

Where $\mathcal{I}$ is the set of all extreme points of the unlinked problem, $\mathcal{P}$, and X is a matrix of all x^i, $i \in \mathcal{I}$. The Lagrangian of $P^{(MP)}$ is

$$\begin{aligned} L(\lambda, q, r) &= c^\top X\lambda + q(b - d^\top X\lambda) + r(1 - \mathbf{1}^\top \lambda), \quad q \geq 0 \\ g(q, r) &= \max_{\lambda \geq 0} L(\lambda, q, r) = qb + r + \max_{\lambda}(c^\top X - qd^\top X - r\mathbf{1}^\top)\lambda \\ &= qb + r + \max_{i \in \mathcal{I}}(c^\top x^i - qd^\top x^i - r) && \text{(since optimum at an extreme point)} \\ &= qb + r + \max_{x \in \mathcal{P}}(c - qd)^\top x - r && \text{(equivalent formulation over } \mathcal{P}) \\ &= qb + \max_{x \in \mathcal{P}}(c - qd)^\top x && \text{(dependence on } r \text{ cancels)} \end{aligned}$$

By construction, the dual function, $g(q,r)$ is an upper bound on $P^{(MP)}$, $\forall q \geq 0$. We can choose simplex multiplier q as the dual for our restricted master problem, and use the solution to our pricing subproblem, $\max_{x\in\mathcal{P}}(c-qd)^\top x$ to get our upper bound:

$$\begin{aligned} qb &= \tfrac{1}{3} * 15 \\ (c-qb)^\top x^3 &= \tfrac{40}{3} \\ g(q,r) &= \tfrac{15}{3} + \tfrac{40}{3} = \mathbf{\frac{55}{3}} \end{aligned}$$

5. **Exercise 6.5 (Given)** Consider an LP of the form

$$\begin{aligned} \min \quad & c^\top x + d^\top y \\ \text{s.t.} \quad & Ax + Dy \leq b \\ & Fx \leq f \\ & y \geq 0 \end{aligned}$$

(a) **(Given)** Access to a fast subroutine for solving $\min_{\{x|Fx\leq f\}} h^\top x$ for arbitrary h. **(Want)** How would I go about solving the problem?

This subroutine lends itself to the Dantzig-Wolfe algorithm. Consider the separable sets

$$\mathcal{P}_x = \{x \mid Fx \leq f\} \qquad\qquad \mathcal{P}_y = \{y \mid y \geq 0\}$$

$\mathcal{P}_x$ may be a complex polyhedron, while $\mathcal{P}_y$ is the nonnegative orthant, trivially composed of extreme rays e_j, $j \in [\text{length}(d)]$. The Dantzig-Wolfe decomposition of the master problem is

$$\begin{aligned} \min \quad & \sum_i \lambda_x^i c^\top x^i + \sum_k \theta_x^k c^\top w_x^k + \sum_j y_j d^\top e_j \\ \text{s.t.} \quad & \sum_i \lambda_x^i Ax^i + \sum_k \theta_x^k Aw_x^k + \sum_j y_j De_j \leq b & (q) \\ & \sum_i \lambda_x^i = 1 & (\text{r}) \\ & \lambda_x^i, \theta_x^k, y_j \geq 0, \ \forall i,j,k \end{aligned}$$

The algorithm then could be:

1. As is done in the textbook, suppose we have a basic feasible solution, $(\tilde{x}, \tilde{y})$, to the master problem and corresponding basis, B.
   - We may find B through the method described in the textbook (p251), which is similar to Phase 1 of the two-phase simplex method.
   - In the subsequent steps, let $B^k \leftarrow B$, $(B^k)^{-1} \leftarrow B^{-1}$
2. Calculate $p^\top = c_{B^k}^\top (B^k)^{-1}$, where c_{B^k} is the vector of objective coefficients corresponding to B^k. Let q be the indexes of p corresponding to the linked constraint, $Ax + Dy \leq b$, and let r be the index corresponding to the convexity constraint
3. Solve the pricing subroutine below. As a given, we have a fast algorithm for this

$$g = \min_x (c^\top - q^\top A)x \quad \text{s.t. } Fx \leq f \qquad\qquad h = (c^\top - q^\top A)$$

   (a) Note by construction of p corresponding to feasible basis B, and inclusion of all e_j in the Dantzig-Wolfe decomposition, we are guaranteed that $d^\top \succeq qD$, so suffice to only check this subproblem.

(b) If $g \geq r$, then all reduced costs in the master problem are nonnegative, and we have an optimal basis, B^k. Terminate with corresponding optimal basic feasible solution.

(c) If $g = -\infty$, then the pricing subroutine problem provides us with an extreme ray of the separable constraint set, w_x^k. The reduced cost of the corresponding variable, θ_x^k, in the Dantzig-Wolfe decomposition of the full master problem is negative and we want to include w_x^k in our restricted master problem.

(d) If $g < r$ but finite, then the pricing subroutine problem provides us with an extreme point of the corresponding separable constraint set, x^i. The reduced cost of corresponding variable, λ_x^k, in the Dantzig-Wolfe decomposition of the full master problem is negative and we want to include λ_x^k in our restricted master problem.

4. Introduce the new column, w_x^k or x^i, and the corresponding variable, θ_x^k, λ_x^i, into the restricted master problem. Use the simplex method with the new column and variable to update $B^k, (B^k)^{-1}$. Return to step 2 and iterate until termination.

(b) **(Given)** Access to a fast subroutine for solving $\min_{\{y|Dy \leq h, y \geq 0\}} d^\top y$ for arbitrary h. **(Want)** How would I go about solving the problem?
This subroutine lends itself to Benders Decomposition (delayed constraint generation algorithm). Consider the reformulation of the provided problem

$$\begin{aligned} \min_{x,z} \quad & c^\top x + z \\ \text{s.t.} \quad & Fx \leq f \\ & p^{i\top}(b - Ax) \leq z, \ \forall i \text{ extreme points of } P = \{p \mid p^\top D \leq d^\top\} \\ & w^{j\top}(b - Ax) \leq 0, \ \forall j \text{ extreme rays of } P \end{aligned}$$

The algorithm then could be

1. Initialize by solving a relaxed problem, perhaps $\min_{\{x,z|Fx \leq f, z \geq z_0\}} c^\top x + z$ and obtain optimal solution $(\tilde{x}^*, \tilde{z}^*)$. In the subsequent steps, let (x^{k*}, z^{k*})
2. Solve the decomposition subproblem, $(P_{x^{k*}})$. As a given, we have a fast algorithm for $(P_{x^{k*}})$

$$\begin{aligned} (P_{x^{k*}}) \quad & \min_y d^\top y && \text{s.t } Dy \leq (b - Ax^{k*}), \ y \geq 0 && h = b - Ax^{k*} \\ (D_{x^{k*}}) \quad & \max_p p^\top (b - Ax^{k*}) && \text{s.t } p \geq 0, \ p^\top D \leq d \end{aligned}$$

(a) If $val(P_{x^{k*}}) \leq z^{k*}$, then all constraints are satisfied and we have an optimal solution to the master problem. Terminate.

(b) If $val(P_{x^{k*}}) > z^{k*}$, then this corresponds to an optimal solution, p^k to $(D_{x^{k*}})$ whose cost is greater than z^{k*}. Include this, $p^{k\top}(b - Ax) \leq z$ as a new constraint in the relaxed master problem

(c) If $(P_{x^{k*}})$ is infeasible, then $(D_{x^{k*}})$ is unbounded and optimal solution to the dual corresponds to extreme ray, w^k. Include this, $w^{k\top}(b - Ax) \leq 0$ as a new constraint in the relaxed master problem.

3. Solve a new relaxed problem of the form below and return to step 2 with optimal (x^{k*}, z^{k*}).

$$\begin{aligned} \min_{x,z} \quad & c^\top x + z \\ \text{s.t.} \quad & Fx \leq f \\ & p^{i\top}(b - Ax) \leq z, \ \forall i = 1, \ldots, k_i \\ & w^{j\top}(b - Ax) \leq 0, \ \forall j = 1, \ldots, k_j \end{aligned}$$

where vectors p^i, w^j are iteratively added to the problem in each step 2 of the iteration.